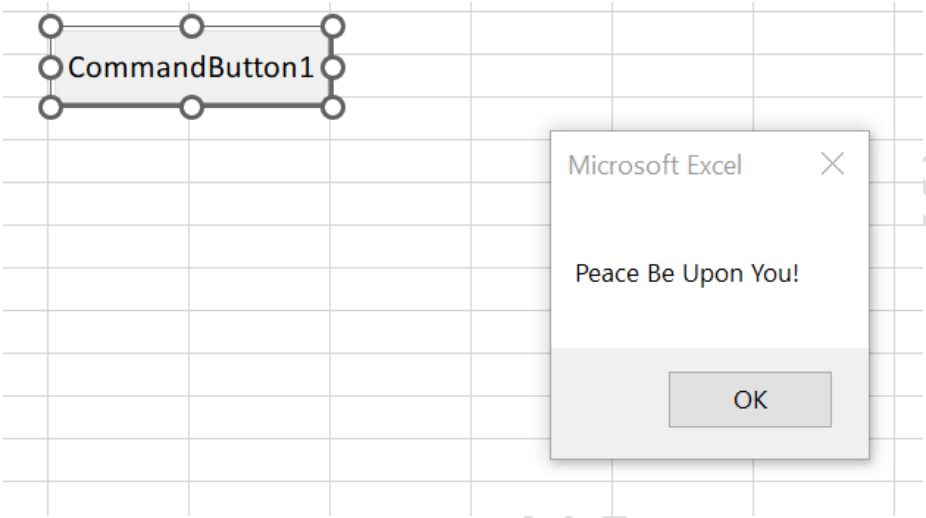
1. Insert a **Command Button** in the **Sheet1** of your **active Workbook**. Assign appropriate **macro** (line/lines of code) so that when a user clicks the **CommandButton1**, a message box will pop up as:



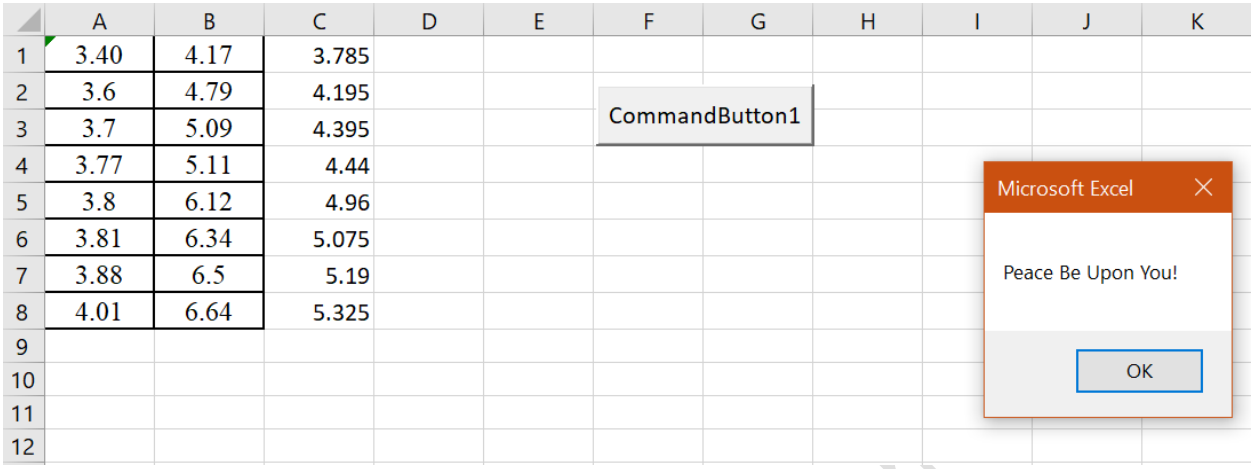
Save your workbook as **VBA.xlsm** file in your computer.

2. Prepare the following data in the same Sheet 1.

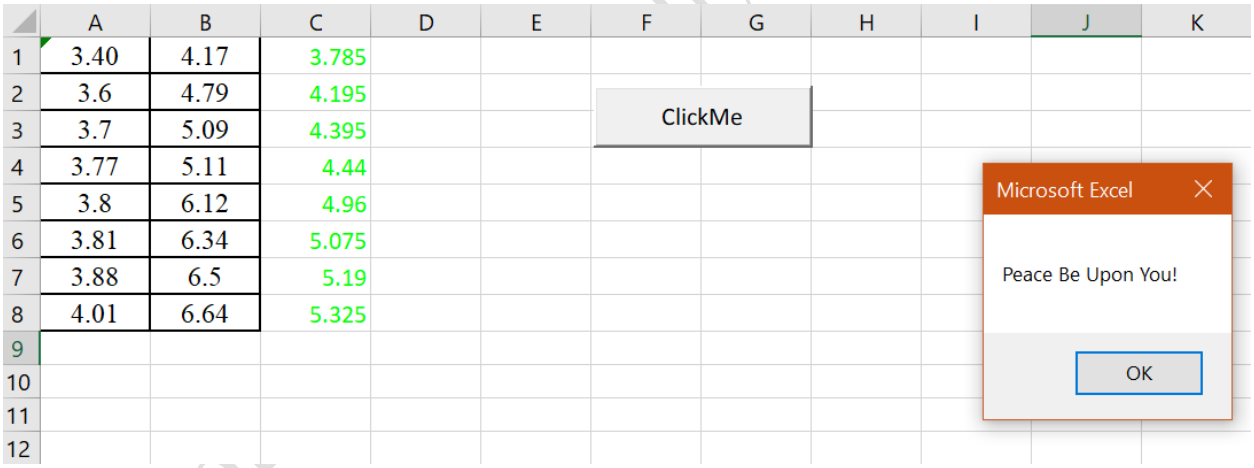
Row	Column A	Column B
1	3.40	4.17
2	3.60	4.79
3	3.70	5.09
4	3.77	5.11
5	3.80	6.12
6	3.81	6.34
7	3.88	6.50
8	4.01	6.64

Using 1 integer variable and single **For-Next-Loop**, add additional macro to the CommandButton1. The objective is to compute the **average** of data in Column A and Column B and to display this in Column C. When a user clicks the CommandButton1, the following output will be shown on the screen.

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3. From Question 2, turn on the **Design Mode** under the **Developer** tab. From the **CommandButton object**, relabel the button as ClickMe. Next, add few suitable macro so that when a user clicks the ClickMe button, the following output will be displayed:



4. Prepare the following data in **Sheet2** of your **active Workbook**.

Row	Column A	Column B
1	- 3.4021	5.0201
2	0.6004	- 0.0206
3	- 5.7009	- 5.0210
4	- 3.7703	1.0217
5	- 3.8032	- 3.0219
6	1.8122	- 4.0222
7	6.8812	6.8812
8	5.0109	4.0225

Insert a new **Command Button** in the **Sheet2**. Using nested **If-Then-Else-End-If** statement, assign appropriate **macro** so that when a user clicks the **CommandButton1**, the following consequences will be reflected in Column C:

a) If data in Column A is greater than data in Column B, the cell value in Column C is TRUE.
b) If data in Column A is smaller than data in Column B, the cell value in Column C is FALSE.
c) If data in Column A is equal to data in Column B, the cell value in Column C is EQUAL.

	A	B	C	D	E	F	G	H	I	J
1	-3.4021	5.0201	FALSE							
2	0.6004	-0.0206	TRUE							
3	-5.7009	-5.021	FALSE							
4	-3.7703	1.0217	FALSE						CommandButton1	
5	-3.8032	-3.0219	FALSE							
6	1.8122	-4.0222	TRUE							
7	6.8812	6.8812	EQUAL							
8	5.0109	4.0225	TRUE							
9										

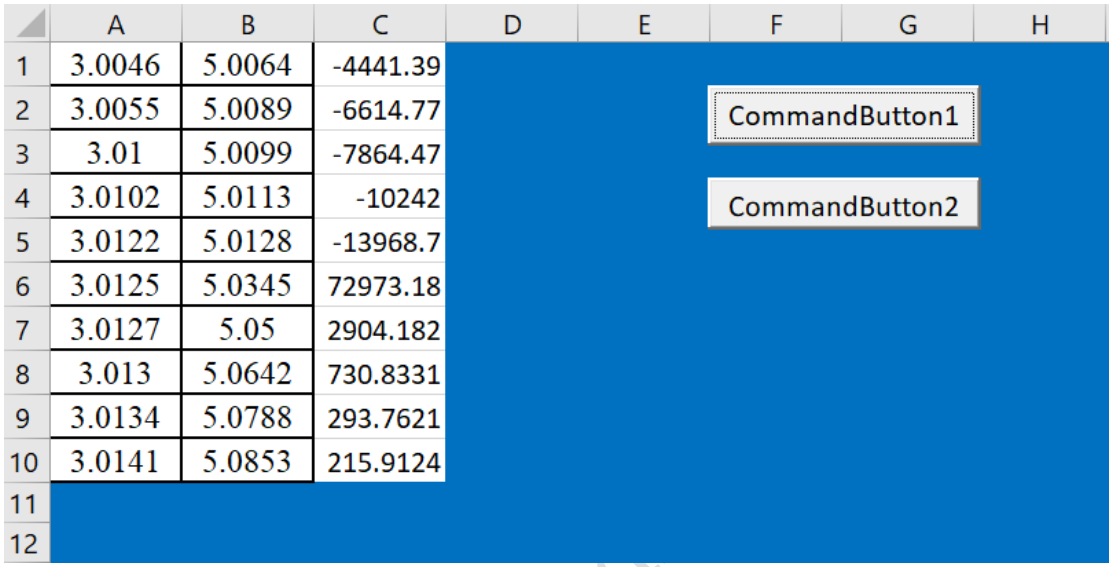
5. Given the function $f(x) = 5\sec^3(5x)\,(4x-1)^{-\frac{1}{3}}$. Use Trapezoidal rule with **Macro** and **Module** to find $\int_a^b f(x)\,dx$ where a (Column A) and b (Column B) represent the lower and upper boundaries of the integral.

Row	Column A	Column B
1	3.0046	5.0064
2	3.0055	5.0089
3	3.0100	5.0099
4	3.0102	5.0113
5	3.0122	5.0128
6	3.0125	5.0345
7	3.0127	5.0500
8	3.0130	5.0642
9	3.0134	5.0788
10	3.0141	5.0853

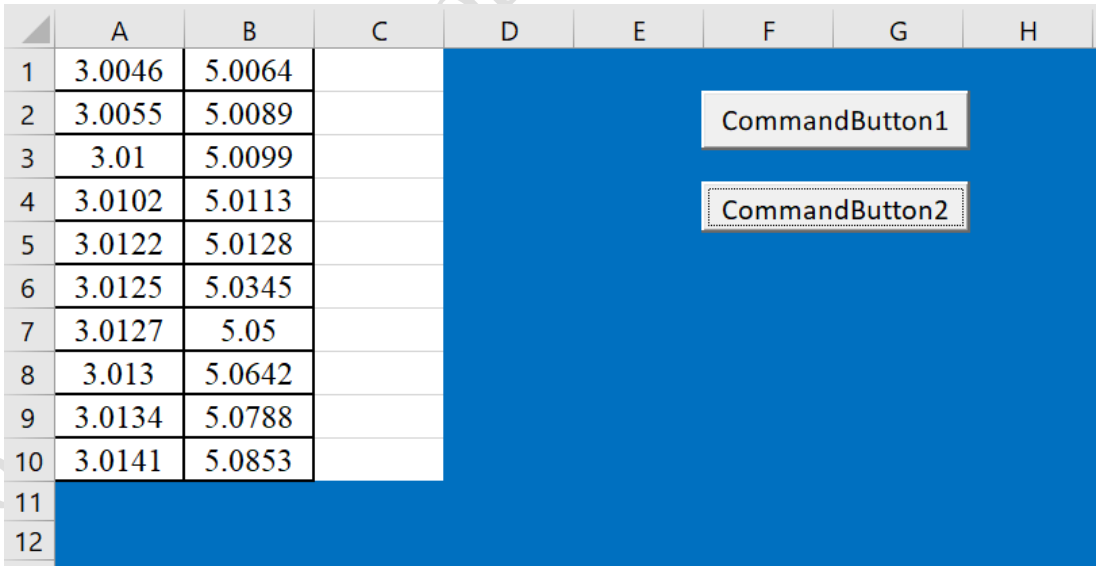
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Your worksheet should have 2 command buttons that each will operate with a single click by a user;

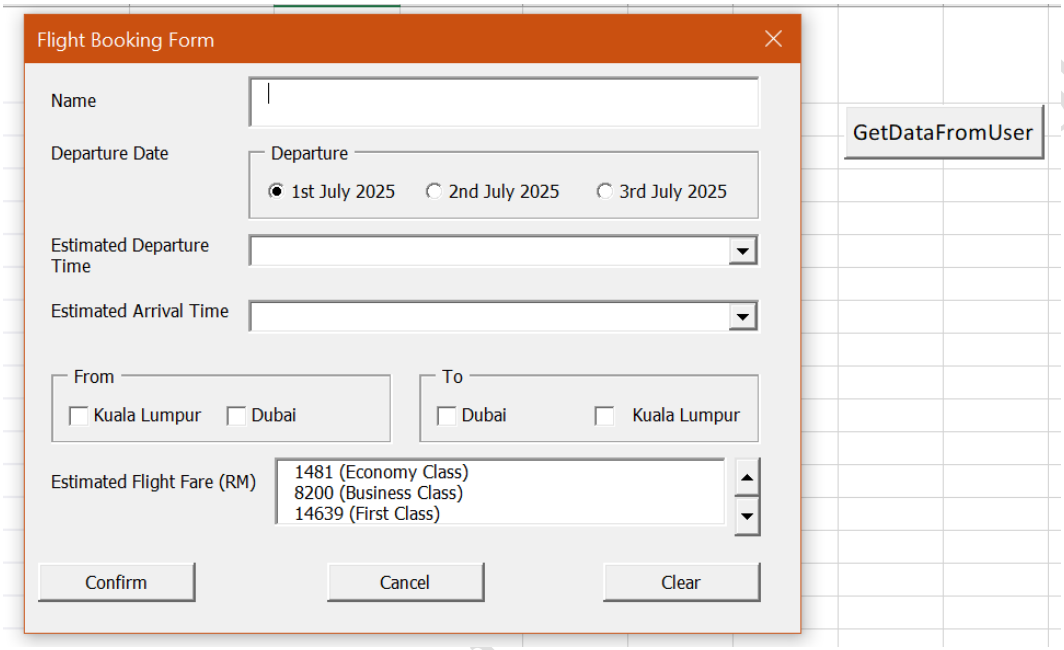
a) **CommandButton1** will show the final integration in Column C

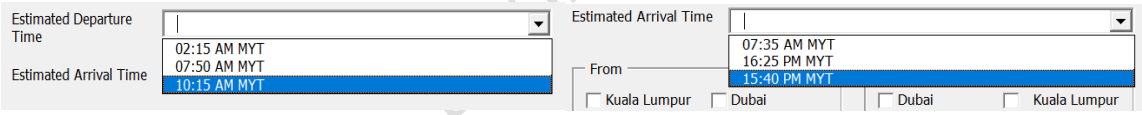


b) **CommandButton2** will clear the contents of all cells in Column C



6. Place a command button in your **Worksheet1** of a Workbook named **VBA2.xlsm**. Rename **CommandButton1** as **GetDataFromUser**. Next, from the VBA, insert and design a **UserForm** to record one-way ticket flight booking information from a user. Upon a click of the GetDataFromUser button, the UserForm will appear as follows:





7. Insert appropriate **Macro** to the **UserForm VBA Editor** so that all the text box, list box, combo boxes, option boxes, check boxes, spin button, Confirm command button, Cancel command button and Clear command button can be executed without errors. Test your UserForm sufficiently.

8. Let a user fill in the UserForm in Question 6 with some variations in the selection. Next, transfer the data from the UserForm to Worksheet1 following the column labels below:

A	B	C	D	E	F	G	H	I	J	K	L	M
Name	**Departure Date**	**Estimated Departure Time**	**Estimated Arrival Time**	**From**	**To**	**Estimated Flight Fare (RM)**					GetDataFromUser	

9. From Question 7, use **Excel** to calculate the flight time duration between the **Estimated Departure Time** and **Estimated Arrival Time** for each row of the data transferred from the UserForm. Show these values in **Column H** under the label **Flight Time Duration**.

10. From Question 8, plot a suitable graph of **Flight Time Duration** versus **Estimated Flight Fare (RM)**.

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